

IDv2 — Technical Report

Interoperable Decentralized Voting v2 (IDv2)

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1. The Problem

Nigeria currently conducts general elections for roughly 93 million registered voters, and projections place that figure beyond 150 million within the next decade. Each election cycle grows more expensive, not only financially but also in terms of public trust.

Polling-unit results are first recorded on paper before being transcribed into state and federal summaries. Every transcription stage creates an opportunity for discrepancies. Voter rolls remain vulnerable to ghost-voter insertion. Vote-buying persists because marked ballots can be photographed and used as proof of compliance. Citizens who register in one location but are elsewhere on election day often lose the practical ability to vote.

A modern electoral system must satisfy four conditions simultaneously. Every counted vote must come from an eligible voter. No voter may be counted twice. The link between voter and ballot must remain unrecoverable. Independent observers must be able to verify the final tally without privileged access.

IDv2 was designed to satisfy those four conditions under the threat model that exists in Nigeria.

2. From IDv1 to IDv2

The earliest design relied on the National Identity Management Commission (NIMC) as a trusted signing authority. Under that model, NIMC would verify eligibility off-chain, sign an attestation, and the Solana program would accept that signature as proof of voting eligibility.

That design was ultimately rejected because the entire system reduced to one question: could the signing authority ever be compromised?

If a signing key were misused, fraudulent eligibility records would become indistinguishable from legitimate ones. The privacy guarantee would also weaken because the signer would know the mapping between voter and credential.

IDv2 removes the signer from the trust model entirely. NIMC remains relevant in the production architecture, but only as a passive attestation source rather than an active signing authority.

3. Why This Matters

IDv2 is not primarily a monetization project. It is civic infrastructure.

Although the system can be adapted for corporate governance and institutional decision-making, its primary purpose is democratic integrity. The central objective is to create societal value by strengthening trust in collective decision-making.

That direction aligns naturally with the cypherpunk principle that individuals should retain control over what belongs to them, especially their opinions and their vote.

4. The IDv2 Solution

A voter registers once at any polling unit using a National Identification Number and biometric verification.

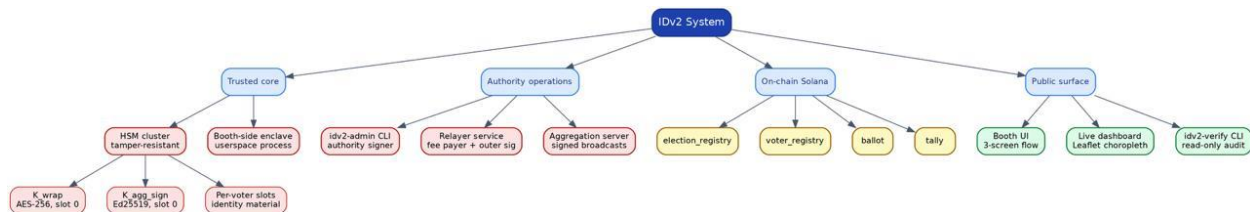
At registration, the system creates a cryptographic commitment proving that the voter exists and is eligible without revealing identity. That commitment is inserted into an on-chain Merkle tree.

Once registered, the voter may appear at any booth in the country. Following biometric authentication, the voter selects a candidate and the vote is submitted to Solana.

No receipt, transaction signature, or portable proof is returned to the voter. This is what makes practical vote-buying verification far more difficult.

The voter's signing material never leaves a hardware security module. It is generated during registration, sealed within the HSM, and invoked only after successful biometric authentication. The voter never directly controls or even possesses a private key.

In a production deployment, NIMC still has a role, but not as a signer. Instead, NIMC functions as a passive attestation source. Given a NIN, it returns a commitment proving that the voter exists in the national register without exposing identifying data.



Place your full architecture organogram here — the diagram beginning with “**IDv2 System**” at the top.

This should appear here because it introduces the complete trust hierarchy immediately after the overall system explanation.

5. Core Security Properties

IDv2 enforces five properties simultaneously.

First, **eligibility**: only voters whose commitments exist in the on-chain Merkle root may produce valid votes.

Second, **one-person-one-vote**: nullifier checks prevent replay or duplicate voting.

Third, **receipt-freeness**: no transaction-identifying artifact is returned to the booth.

Fourth, **integrity**: the HSM-backed signature and Solana consensus ensure that votes cannot be forged or altered after submission.

Fifth, **public verifiability**: every valid ballot exists as a public on-chain record that can be independently audited.

6. Registration Procedure

Registration begins when a voter appears at a registration unit.

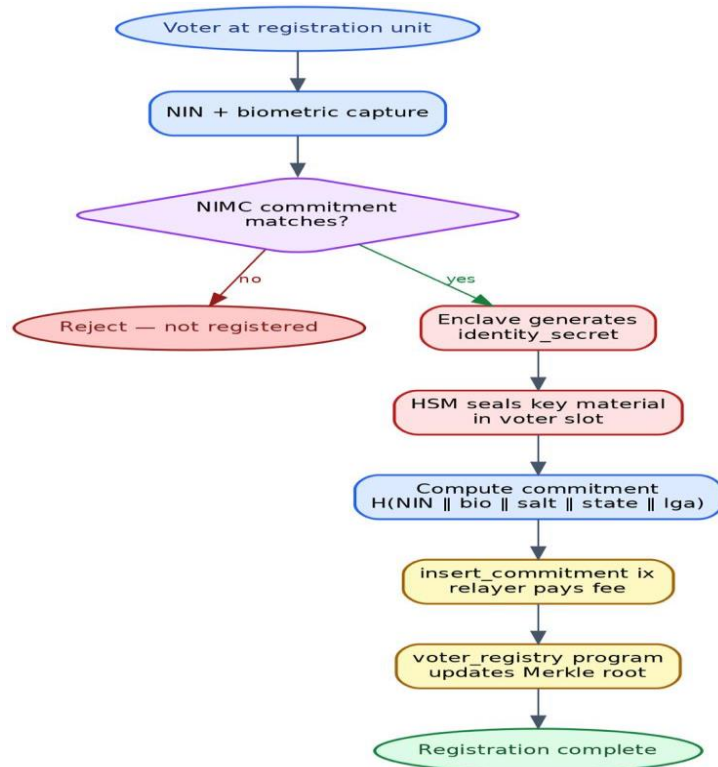
The voter presents a NIN and completes biometric capture. The booth verifies that the NIMC commitment matches the expected commitment derived from the registration inputs.

If verification fails, registration is rejected.

If verification succeeds, the enclave generates the voter's identity secret. The HSM seals the signing material inside the voter slot. A commitment is then computed using the voter's identifying inputs and jurisdiction metadata.

The commitment is submitted on-chain through the relayer. The voter registry updates the Merkle root. Once the root is updated, registration is complete.

At no point does the voter's signing material leave the HSM. Only the commitment crosses the trust boundary.



Place the vertical registration flow diagram here — the one beginning with “**Voter at registration unit.**”

This is the right place because the paragraph directly describes that exact sequence.

7. Ballot Cast Procedure

A registered voter may appear at any authorized booth.

The voter authenticates with fingerprint and NIN. If biometric verification fails, the attempt is rejected.

If verification succeeds, the booth displays the candidate list. The voter selects a candidate.

The enclave constructs the vote message together with the Merkle proof and nullifier. The system then checks that the nullifier has not been used before.

If the nullifier is invalid or already consumed, the vote is rejected.

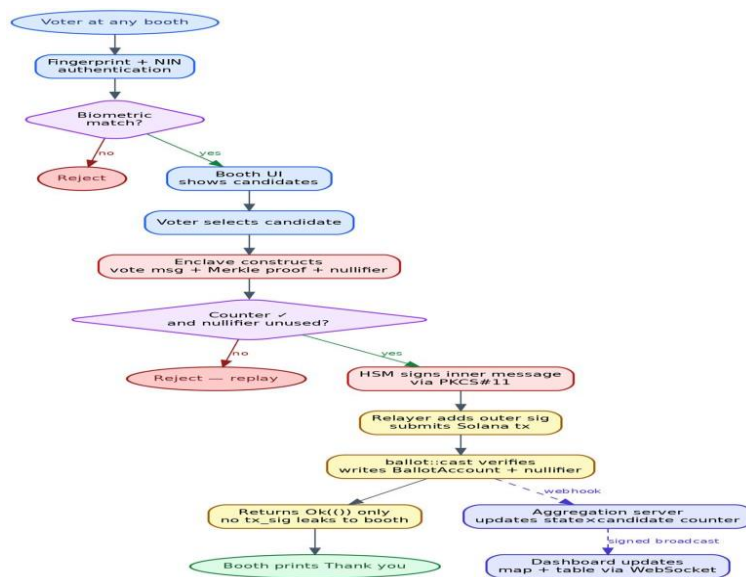
If valid, the HSM signs the inner vote message. The relay adds the outer signature and submits the transaction to Solana.

The ballot program verifies the proof, records the ballot, and stores the nullifier.

The booth receives only a simple success acknowledgment. No transaction signature is returned.

Separately, the aggregation server updates the live dashboard through signed broadcast.

This separation between the booth and the public tally path is central to receipt-freeness.



Place the vertical vote-cast flow diagram here — the one beginning with **“Voter at any booth.”**

This belongs here because the preceding text explains the entire hot-path voting sequence.

8. Scale and Deployment

The voter registry is implemented as a Merkle tree of depth 28, supporting approximately 268 million slots.

This comfortably covers Nigeria's projected voter population while leaving room for expansion.

To reduce recurring costs, IDv2 includes a rent-recovery mechanism. After an election is finalized, per-vote records can be closed and the rent recycled for future elections.

This transforms locked capital into reusable infrastructure rather than recurring cost.

A successful deployment would demonstrate that Solana infrastructure can support not only financial applications but also civic verification and public governance at population scale.

9. Practical Limitations

IDv2 closes much of the post-registration attack surface, but it does not solve every institutional problem.

Registration still depends on physical polling units under state-level coordination. If a demographic is never registered, the blockchain cannot detect the omission.

The current demo also uses SoftHSM2 rather than production HSM hardware. The cryptographic surface remains the same, but the physical tamper-resistance properties of production hardware are not exercised.

The NIMC integration is mocked in the demo. A production deployment would require institutional cooperation.

These are operational and political constraints rather than architectural flaws.

10. Closing

IDv2 is a complete cryptographic voting system designed around Nigerian electoral realities and implemented on Solana infrastructure.

The remaining work is not purely technical. Institutional adoption, operational integration, and pilot deployment require stakeholders beyond the reach of a solo developer.

What this submission demonstrates is that the technical foundation already exists.

The technology is ready when the politics is.